

Rotations:

$$R_B^A = \begin{pmatrix} x_B^A & y_B^A & z_B^A \end{pmatrix}$$

The rotation matrix

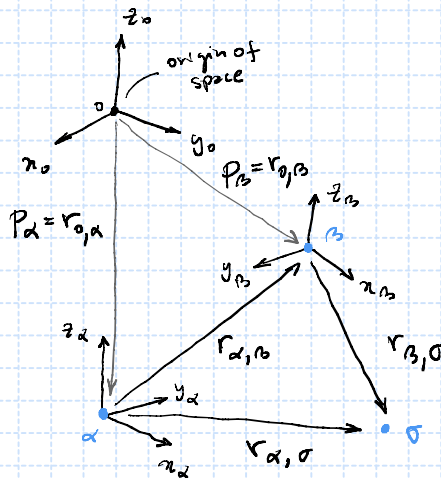
$$r^A = R_B^A r^B$$

Transformation of a vector from a frame into another frame

$$R_B^A = R_B^C R_C^A$$

Composition of rotation matrices

Translations:



$$r_{A,S} = r_{A,B} + r_{B,S}$$

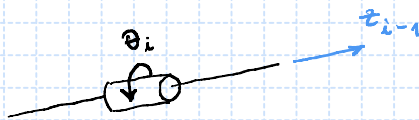
$$r_{A,S}^S = r_{A,B}^S + R_B^S r_{B,S}^S$$

homogeneous transformations

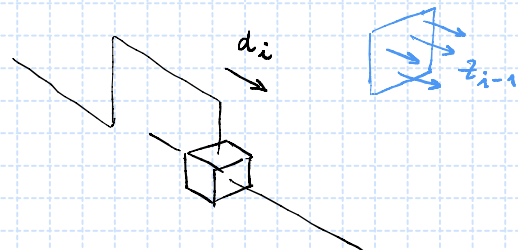
$$r_{A,S} = -r_{S,A}$$

$$r_{A,S} = P_S - P_A$$

Joint diagrams (Diagramas de junta)



A revolute joint i defines the axis of rotation z_{i-1}



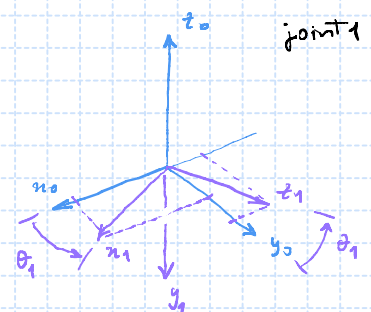
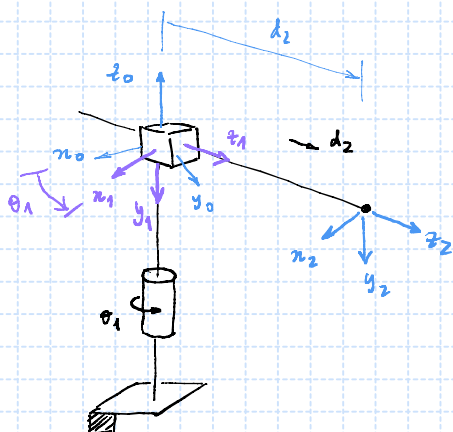
A prismatic joint i defines a displacement field z_{i-1}

1.

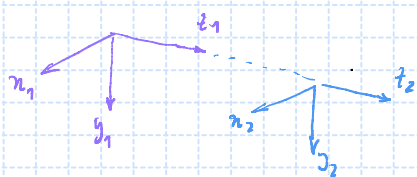
direct kinematics: $r_{0,2}$, R_2^0

• Rotations:

$$\begin{aligned} R_1^0 &= \begin{pmatrix} x_1^0 & y_1^0 & z_1^0 \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta_1 & 0 & -\sin \theta_1 \\ \sin \theta_1 & 0 & \cos \theta_1 \\ 0 & -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} c_1 & 0 & -s_1 \\ s_1 & 0 & c_1 \\ 0 & -1 & 0 \end{pmatrix} \end{aligned}$$



joint 2



$$R_2^1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

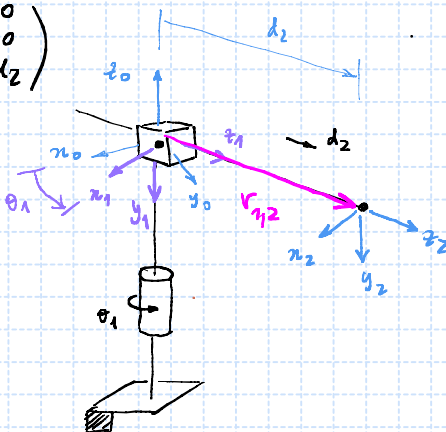
$$R_2^0 = R_1^0 R_2^1 = R_1^0 I = \begin{pmatrix} c_1 & 0 & -s_1 \\ s_1 & 0 & c_1 \\ 0 & -1 & 0 \end{pmatrix}$$

• translations:

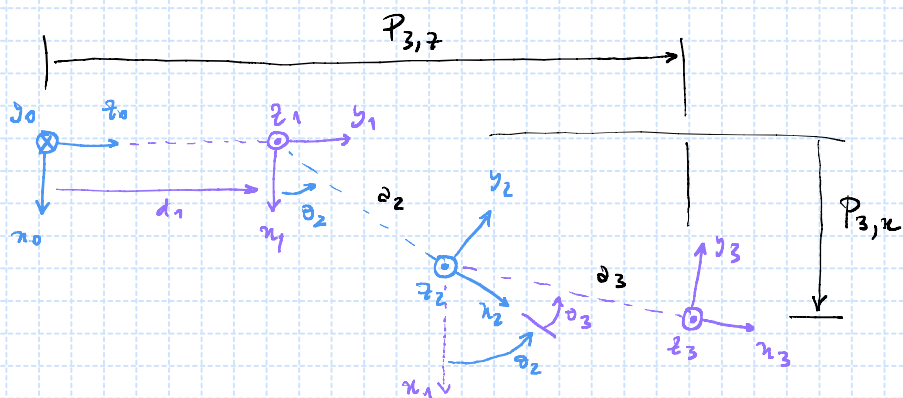
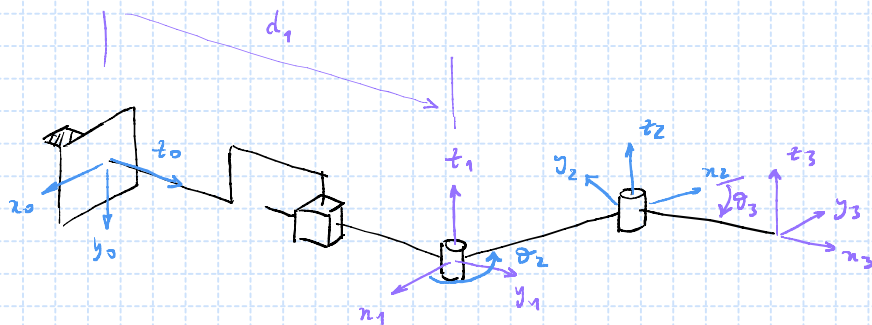
$$P_2 = r_{0,2}^0 = R_1^0 r_{0,1}^1 + R_2^0 r_{1,2}^2$$

$$P_2 = \begin{pmatrix} c_1 & 0 & -s_1 \\ s_1 & 0 & c_1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ d_2 \end{pmatrix} = \begin{pmatrix} -d_2 s_1 \\ d_2 c_1 \\ 0 \end{pmatrix}$$

$$r_{1,2}^2 = \begin{pmatrix} 0 \\ 0 \\ d_2 \end{pmatrix}$$



2.



$$P_3 = \begin{pmatrix} P_{3,x} \\ P_{3,y} \\ P_{3,z} \end{pmatrix} = \begin{pmatrix} P_3 \cdot \hat{i} \\ P_3 \cdot \hat{j} \\ P_3 \cdot \hat{k} \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (\hat{i} \hat{j} \hat{k}) = R_\alpha^\alpha$$

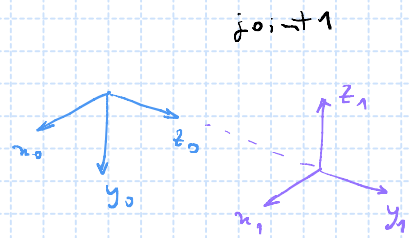
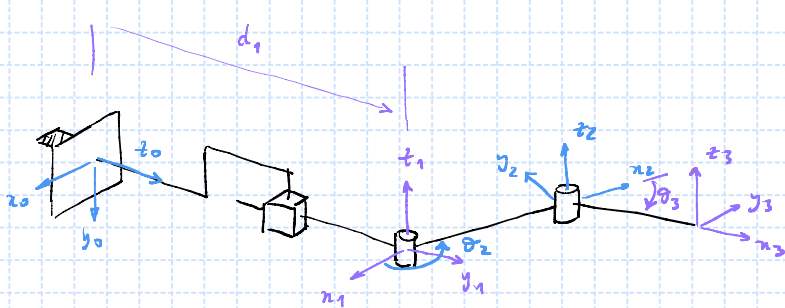
$$P_{3,z} = d_1 + a_2 s_2 + a_3 s_{23}$$

$$P_{3,x} = a_2 c_2 + a_3 c_{23}$$

$$P_3 = \begin{pmatrix} a_2 c_2 + a_3 c_{23} \\ 0 \\ d_1 + a_2 s_2 + a_3 s_{23} \end{pmatrix}$$

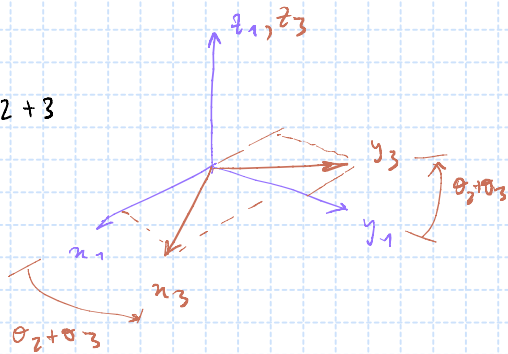
$$s_{23} = \sin(\theta_2 + \theta_3)$$

$$c_{23} = \cos(\theta_2 + \theta_3)$$



$$R_1^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

joints 2+3



$$R_3^1 = \begin{pmatrix} \cos(\theta_2 + \theta_3) & -\sin(\theta_2 + \theta_3) & 0 \\ \sin(\theta_2 + \theta_3) & \cos(\theta_2 + \theta_3) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_3^0 = R_1^0 R_3^1 = \begin{pmatrix} c_{23} & -s_{23} & 0 \\ 0 & 0 & -1 \\ s_{23} & c_{23} & 0 \end{pmatrix}.$$

direct kinematics:

$$P_3 = \begin{pmatrix} a_2 c_2 + a_3 c_{23} \\ 0 \\ d_1 + a_2 s_2 + a_3 s_{23} \end{pmatrix}, \quad R_3^0 = \begin{pmatrix} c_{23} & -s_{23} & 0 \\ 0 & 0 & -1 \\ s_{23} & c_{23} & 0 \end{pmatrix}.$$